GitHub Codespaces Debugging & Workspace Setup

Portfolio Case Study & Technical Teaching Guide: Machine Learning Environment Configuration

PROJECT REPO	ENVIRONMENT	PYTHON VERSION
Modeling-with-python	**GitHub Codespaces (Linux)**	**3.14.2 Runtime**

1. Executive Summary

This technical case study documents the complete setup, troubleshooting, and execution process for a Machine Learning pipeline inside a GitHub Codespace environment. During initial project initialization, multiple real-world syntax, pathing, and dependency resolution errors were identified and resolved step-by-step.

2. Phase 1: File Path & Extension Resolution

Issue Encountered: FileNotFoundError & Syntax Errors

The initial execution failed due to typographical errors in the dataset filename extension (`.cvs` vs `.csv`) and function naming (`read_cvs` vs `read_csv`), as well as mismatched string quotes in terminal print statements.

```
# Incorrect Syntax (Initial State)
df = pd.read_cvs('Placement_Data_Full.cvs')
print('\n---Dataset Info---")  # Mismatched quotes

# Corrected Implementation
df = pd.read_csv('Placement_Data_Full.csv')
print("\n--- Dataset Info ---")
```

Root Cause & Fix

1. Renamed repository asset to exact match `Placement_Data_Full.csv`.
2. Updated Pandas invocation from `read_cvs()` to standard `read_csv()`.
3. Balanced string literal quotes on line 7 to remove `SyntaxError: unterminated string literal`.

3. Phase 2: Dependency Resolution & Environment Package Installation

When adding advanced Machine Learning imports (`scikit-learn`, `matplotlib`, `seaborn`), the system raised environment import failures.

> **Issue Encountered: ModuleNotFoundError & Pip Lookup Failure**
>
> 1. `ModuleNotFoundError: No module named 'pandas'`
> 2. `ERROR: Could not find a version that satisfies the requirement matplotlib` due to Python 3.14 preview environment constraints and outdated package manager indices.

4. Step-by-Step Resolution Workflow

1. **Upgrade Package Manager:** Refreshed pip index within the active Linux virtual environment.

```
python3 -m pip install --upgrade pip
```

2. **Isolated Package Installation:** Installed core machine learning dependencies individually to ensure wheel compatibility.

```
pip install pandas
pip install scikit-learn
pip install seaborn matplotlib
```

5. Final Verification & Data Execution Results

Upon completing dependency updates, executing `python main.py` validated that all 215 dataset rows were correctly parsed and ingested into memory without missing schema definitions.

Column Name	Non-Null Count	Data Type	Status
`sl_no`	215 non-null	int64	Loaded
`gender`, `ssc_b`, `hsc_b`, `hsc_s`	215 non-null	string (object)	Loaded
`ssc_p`, `hsc_p`, `degree_p`, `etest_p`, `mba_p`	215 non-null	float64	Loaded
`status` (Target Variable)	215 non-null	string (Placed / Not Placed)	Ready for ML
`salary`	148 non-null	float64	Identified (67 NaN)